ZenVoice vs FluidVoice Gap Analysis Report

Executive Summary

FluidVoice is a mature, GPLv3 open-source macOS dictation app with 9.3k GitHub stars. It ships a wide model catalog, real-time live preview with notch support, command/rewrite modes, optional cloud AI providers, analytics, and automatic updates. ZenVoice is currently a private-source macOS dictation app at v0.2.0 with strong privacy engineering (encrypted history, no accounts, no cloud). The user's goal is to pivot ZenVoice into a **fully free, open-source, developer-first** dictation product that competes with FluidVoice while keeping a stronger multilingual focus (not English-first). This report highlights what ZenVoice is missing, where FluidVoice has weaknesses ZenVoice can exploit, and recommended feature priorities. ---

1. What ZenVoice Currently Has (Strengths)

- Native macOS menu-bar app with global hotkey and hold-to-dictate
- Encrypted local transcript history (AES-GCM + Keychain)
- Private local insights / stats
- Verified model catalog with SHA-256 validation
- Parakeet/CoreML + whisper.cpp runtimes
- 64 language profiles, Hinglish Latin/native/translation modes
- Deterministic Instant Refine (filler/repetition cleanup + meaning guard)
- Per-application language/refinement profiles
- Memory-only context box
- Recovery Inbox for failed/partial dictations
- First-run onboarding
- M9 security review + release-readiness framework
- No accounts, no analytics, no cloud transcription

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2. Major Gaps vs FluidVoice

2.1 Model & Runtime Diversity

	Capability	FluidVoice	ZenVoice	ZenVoice Gap	
	Parakeet TDT v3	Yes	Yes	—	
	Parakeet TDT v2 / Flash	Yes	Partial (Parakeet Unified English only)	Add Parakeet Flash/v2 specialized models	
	Nemotron Speech 3.5 (streaming + offline)	Yes	No	Add Nemotron for fast multilingual streaming	
	Cohere Transcribe	Yes	No	Add Cohere CoreML path for 14-language accuracy	
	Apple Speech (legacy + analyzer)	Yes	No	Add zero-download native macOS fallback	
	Whisper tiny/base/small/medium/large/turbo	Yes (transcribe.cpp GGUF)	Yes (whisper.cpp XCFramework)	Consider GGUF/quantized Whisper for smaller downloads	
	Qwen3 ASR	Hidden/beta	No	Optional future exploration	
	Streaming real-time preview	Yes, fast	Stable partial preview exists but not as polished	Improve streaming latency and UI	

2.2 User Interface / Experience

	Capability	FluidVoice	ZenVoice	Gap	
	Notch-aware live overlay	Yes	No (ZenBar is bottom-edge only)	Add notch/top overlay option	
	Configurable overlay sizes	Yes (pill/large)	No	Add size presets	
	Command Mode (agentic Mac control)	Yes	No	Add voice commands for system actions	

Rewrite Mode (selected-text AI rewrite)	Yes	No	Add inline rewrite/improve mode
Real-time word appearance as you speak	Yes ("insanely fast Parakeet")	Partial	Optimize streaming pipeline
Today-usage stats toolbar pill	Yes	Yes (Insights exists)	Surface stats in ZenBar/menu bar
Meeting / file transcription	Yes	No	Add file/meeting transcription
Custom themes / light-dark compact switcher	Yes	In progress (uncommitted redesign)	Complete redesign, add theme toggle

2.3 AI Enhancement Layer

Capability	**FluidVoice**	**ZenVoice**	**Gap**
Local on-device enhancement model ("Fluid Intelligence")	Private/proprietary runtime	No	ZenVoice deliberately removed generative refinement; decide if a fully open local model is wanted
Cloud AI providers (OpenAI, Groq, custom)	Yes	No	Optional bring-your-own-key enhancement
Per-app prompt sets	Yes	Per-app profiles exist but simpler	Add prompt-based per-app routing
Smart formatting / context-aware capitalization	Yes (via Fluid Intelligence or cloud)	Instant Refine is deterministic	Add optional local or BYO-cloud enhancement

2.4 Distribution & Open Source

Capability	**FluidVoice**	**ZenVoice**	**Gap**
Open source license	GPLv3	Proprietary source-visible	Must relicense to an OSI-approved license (e.g. MIT/Apache/GPLv3)
Homebrew cask	Yes	No	Add brew cask after public release
Auto-updater (AppUpdater)	Yes	No	Add Sparkle or AppUpdater

	GitHub Sponsors / community	Yes	No	Build public repo, issues, Discord	
	Public release downloads	Yes	Private beta only	Complete notarization + clean-device QA	
	Analytics (opt-in)	PostHog, enabled by default	None	Decide whether anonymous opt-in analytics are acceptable	

2.5 Engineering / Code Quality

	Capability	**FluidVoice**	**ZenVoice**	**Gap**	
	Deterministic executable checks (core/storage/runtime)	Limited (xcodebuild tests only)	Yes (3 check executables)	ZenVoice is stronger here	
	Architecture	Single app target (~77k LOC)	Modular (ZenVoice/Storage/Core/Runtime)	ZenVoice is cleaner for contributors	
	Swift Package Manager only	Yes	Yes	Both good	
	Minimum macOS	15.0	14.0	ZenVoice supports older Macs	
	Model provenance / checksum manifest	Partial	Stronger (SHA-256 + pinned revisions)	ZenVoice is stronger	
	Security review doc	Not visible	M9 review exists	ZenVoice stronger	
	Encrypted transcript storage	Keychain for API keys only; history likely unencrypted	AES-GCM + Keychain	ZenVoice is far stronger on privacy	

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3. FluidVoice Weaknesses ZenVoice Can Exploit

1. **Privacy posture is weaker**: FluidVoice collects opt-in analytics by default, has optional cloud AI providers, and its "Fluid Intelligence" local model is a closed/private runtime. ZenVoice can differentiate as **100% local, no telemetry, no cloud, no closed runtimes**. 2. **No encrypted transcript history**: FluidVoice stores

transcription history; ZenVoice encrypts it at rest. 3. **Closed enhancement runtime**: "Fluid Intelligence" is privately maintained. ZenVoice can either skip generative enhancement or use fully open models. 4. **English-first marketing**: Parakeet Flash/TDT v2 and Nemotron are promoted for English/low-latency. ZenVoice can lead with **multilingual-first onboarding** and Hinglish/indic support. 5. **Single massive target**: ~77k LOC in one app target makes contributor onboarding harder. ZenVoice's modular targets are easier to extend. 6. **GPLv3 copyleft**: Some commercial/enterprise users avoid GPLv3. ZenVoice could choose MIT/Apache for broader adoption. 7. **Analytics default-on**: Privacy-conscious developers dislike opt-out analytics. ZenVoice can default-off or omit entirely. ---

4. Recommended ZenVoice Feature Roadmap (Free, Open Source)

Phase 1: Open-source foundation (must do before competing)

- Pick open-source license (MIT/Apache recommended for developer adoption; GPLv3 if copyleft is intentional)
- Make GitHub repo public, add CONTRIBUTING.md, issue templates, CI
- Add README quick-start with Homebrew path (future)
- Remove proprietary license and closed branding decisions

Phase 2: Catch up to core dictation parity

- Add Nemotron Speech 3.5 streaming/offline model path (fast multilingual)
- Add Cohere Transcribe CoreML path (high-accuracy 14-language)
- Add Apple Speech Analyzer fallback (zero-download, macOS 26+)
- Add Parakeet Flash/TDT v2 specialized English models
- Add real-time live preview notch/top overlay option
- Improve streaming transcription latency to match FluidVoice

Phase 3: Differentiating features

- **Multilingual-first onboarding**: language selection before model selection
- **Hinglish + Indic-first improvements**: better transliteration, code-mixed handling
- **Encrypted local transcription history** (already done; market it)
- **Open local enhancement**: use fully open small LLM (e.g. Qwen2.5/Phi) for optional deterministic formatting, or skip enhancement entirely

- **Command mode**: simple deterministic voice commands (no LLM required)
- **Rewrite mode**: optional selected-text rewrite using local or BYO-cloud provider
- **Meeting/file transcription**: drag-and-drop audio/video transcription
- **Plugin/script extensibility**: let developers add custom post-processing scripts

Phase 4: Distribution & community

- Notarized public beta/release
- Homebrew cask
- Auto-updater (Sparkle/AppUpdater)
- Public analytics opt-in (or none)
- Discord / GitHub Discussions

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5. Model Strategy Recommendation

For a free multilingual-focused dictation app, prioritize: 1. **Nemotron Speech 3.5 Streaming** — fastest multilingual streaming, ~40 languages, ~670 MB 2. **Nemotron 3.5 Offline** — higher accuracy multilingual, ~530 MB 3. **Parakeet TDT v3** — fast European/multilingual, ~500 MB 4. **Cohere Transcribe** — high-accuracy 14-language, ~1.4 GB 5. **Whisper Small/Large Turbo** — broad 99-language fallback, GGUF for smaller size 6. **Apple Speech Analyzer** — zero-download native fallback for macOS 26+ Keep English models (Parakeet v2/Flash) available but not the default. ---

6. Risk Assessment

	Risk	Level	Mitigation	
	Relicensing ZenVoice from proprietary	Medium	Need author consent; replace LICENSE; check no third-party GPL contamination	
	FluidAudio dependency is closed-source / branch-based	Medium	FluidVoice uses branch-based FluidAudio; ZenVoice should pin exact revisions or use open alternatives	
	Model download bandwidth/cost	Medium	Use Hugging Face direct downloads; no central server	

	Maintaining multiple ASR backends	High	Start with 2–3 backends, add incrementally	
	GPLv3 if copying FluidVoice code	High	Do not copy FluidVoice code directly; design independently to avoid license contamination	
	macOS 14 vs 15 minimum	Low	Keep macOS 14 target for broader reach	

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7. Conclusion

ZenVoice already has a stronger privacy and modular architecture foundation than FluidVoice. Its main gaps are:

- Open-source licensing and public community
- Model diversity (especially Nemotron, Cohere, Apple Speech)
- Real-time streaming UX and overlay design
- Command/rewrite/meeting modes
- Distribution channels (Homebrew, auto-updater)

The biggest competitive advantage ZenVoice can build is **"FluidVoice but fully open, privacy-first, and multilingual-first"**. If ZenVoice keeps every byte local by default, avoids closed runtimes, and leads with Hinglish/Indic/multilingual quality, it can carve out a distinct developer audience.